

What Is New in Version 2.0

A major update from the publicly released Workbench v1.0.0

Version 2.0 keeps the familiar path from model inputs to completed comparisons while adding safer coordination for large studies, clearer analysis choices, predictable exports, and a signed and notarized Apple Silicon distribution.

Complete Hypercube workflow

Hypercube is now its own fifth tab. Build defines up to two bounded parameter axes, Review confirms scope and resources, Run completes only missing or failed work, Analyze provides cached response matrices and scenario rankings, and Export creates external CSV packages without changing Datastore-only retention. Run status includes a measured ETA, stable completed elapsed time, scoped stopping, and one completion notification for the project.

Find All Changes and cached analysis

Find All Changes scans every eligible output and completed Hypercube case without requiring a single-variable calculation first. It has independent Year and Aggregation controls with Median as the default, searchable Outputs and Scenarios rankings, accessible explanations, case-ranking drilldowns, progress, cancellation, notifications, and fingerprinted cache reuse. Response Matrix requests are cached separately and cell selection is immediate.

Stronger scenario and project tools

Create has clearer regional-package, project, single-file, and Batch Change tools. Scenario names are unique without regard to capitalization. Copies preserve the source name when possible and resolve collisions as Name (copy), Name (copy 2), and so forth while keeping copied result provenance aligned.

Coordinated model execution

Normal and Hypercube work share one FIFO batch queue so unrelated batches never compete for runtime slots. Standard jobs remain individually visible, while each Hypercube becomes one compact Run-history card. Stop controls verify container ownership, duplicate backends are prevented, and exit 137 is called memory-related only when Docker confirms OOMKilled.

Redesigned comparison and exports

Each Compare view keeps independent result selectors and state. Find All Changes searches selected results without depending on the detailed table. Maps and charts preserve completed work while updates load. Every user export uses one FIFO queue; Hypercube Export can package the baseline or up to three completed cases as separate ZIPs with temporary output CSV generation and cleanup.

Improved maps and regional data

Maps use the colors chosen in Settings, adapt scales to visible values, and preserve signs, exact zeros, and unavailable percentages. Fit Project leaves useful context around the study area, supported Marea views show clean selectable regions, and Virginia source data can build MPO or custom regional packages without implying a supported statewide run.

Signed distribution and clearer safeguards

Settings provides package previews, runtime verification, update checks, memory guidance, diagnostics, notifications, and clear storage categories. Authoritative Datastores stay separate from disposable comparison and Hypercube caches. The Apple Silicon application uses Developer ID signing, hardened runtime, Apple notarization, and a stapled ticket for normal macOS installation.

Release scope

This overview covers the Apple Silicon macOS application. Intel macOS and Windows remain separate implementations. Hypercube time and resource estimates are planning guidance rather than guarantees, and a complete statewide Virginia model run is not supported in Version 2.0.

Comparison baseline Public Workbench v1.0.0 release | Version 2.0 signed and notarized edition | September 2026